

Analysis of Clayton Kershaw's Pitch Performance (2015-2019)

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Introduction

Baseball, often referred to as America's favorite pastime, has been respected as one of America's most difficult sports since the 1800's. The MLB, Major League Baseball, is the highest level of baseball in America which consists of 30 teams who are the top 750 baseball players in the nation. Furthermore, this garners the attention of millions of fans young and old, inspiring kids of the future to play this 150 year old game. Aside from competing for sport and entertainment, baseball offers much in the statistical world, providing a massive amount of data for game strategies or player analysis.

One of the most critical positions in baseball is the starting pitcher. Many highly respected players played this position such as Pedro Martinez, Greg Maddux, Randy Johnson, etc. In this project we aim to analyze the performance of starting pitcher Clayton Kershaw of the Los Angeles Dodgers from the years 2015 to 2019. We will examine the rate at which he throws certain pitches and at what release velocity to understand Kershaw's performance spread over the time span. We will also aim to investigate the effectiveness of each of his pitches, e.g., four seam fastball, curveball, and slider, due to their differences in release speed and vertical or horizontal movement.

Overall, we present our findings, detailing the methods used, results obtained, and implications of our analysis. Utilizing data from the MLB Statcast database and provided R package, various statistical techniques and visualizations were applied to understand the evolution of his pitching performance and effectiveness over 2015-2019.

Data Collection

The data for this analysis was collected using the `baseballr` package in R. Clayton Kershaw's player ID was retrieved and used to gather detailed pitch data for the years 2015 to 2019. The key variables included pitch type, release speed, launch speed, spin rate, and pitch movement.

Data Processing

The collected data was processed to filter relevant information and create specific datasets for analysis:

1. **All pitches:** Combined data for all pitches thrown by Kershaw.
2. **Strikes:** Data filtered to include only pitches that resulted in strikes.
3. **Balls in play:** Data filtered to include only pitches that resulted in balls being put into play.

Summary Statistics

Release Speed

- **Overall:** The average release speed across all years was 87.86 mph, with a maximum of 97.60 mph and a minimum of 46.20 mph.
- **By Year:**
 - 2015: Mean = 89.8 mph, Median = 89.8 mph
 - 2016: Mean = 89.8 mph, Median = 89.8 mph
 - 2017: Mean = 88.2 mph, Median = 89.8 mph
 - 2018: Mean = 88.5 mph, Median = 90.1 mph
 - 2019: Mean = 88.5 mph, Median = 89.8 mph

Pitch Type Distribution

- **Four-Seam Fastball (FF):** Most frequently used pitch.
- **Slider (SL):** Second most frequently used pitch.
- **Curveball (CU):** Regularly used pitch.
- **Changeup (CH), Sinker (SI), and Intentional Ball (IN):** Less frequently used.

Pitch Performance Analysis

Release Speed Distribution

- The distribution of release speeds showed a slight decrease over the years, indicating a potential decline in pitch speed as Kershaw's career progressed. This trend is particularly noticeable in the Four-Seam Fastball (FF) pitches.

Pitch Movement

- **Slider (SL):** Horizontal movement remained relatively consistent across the years, with the highest variability observed in 2015. This consistency suggests that Kershaw maintained effective control over his slider.
- **Curveball (CU):** Vertical movement showed a decline from 2015 to 2019, with the most significant drop observed between 2016 and 2017. This decrease could indicate a reduction in the effectiveness of his curveball over time.

Methodology

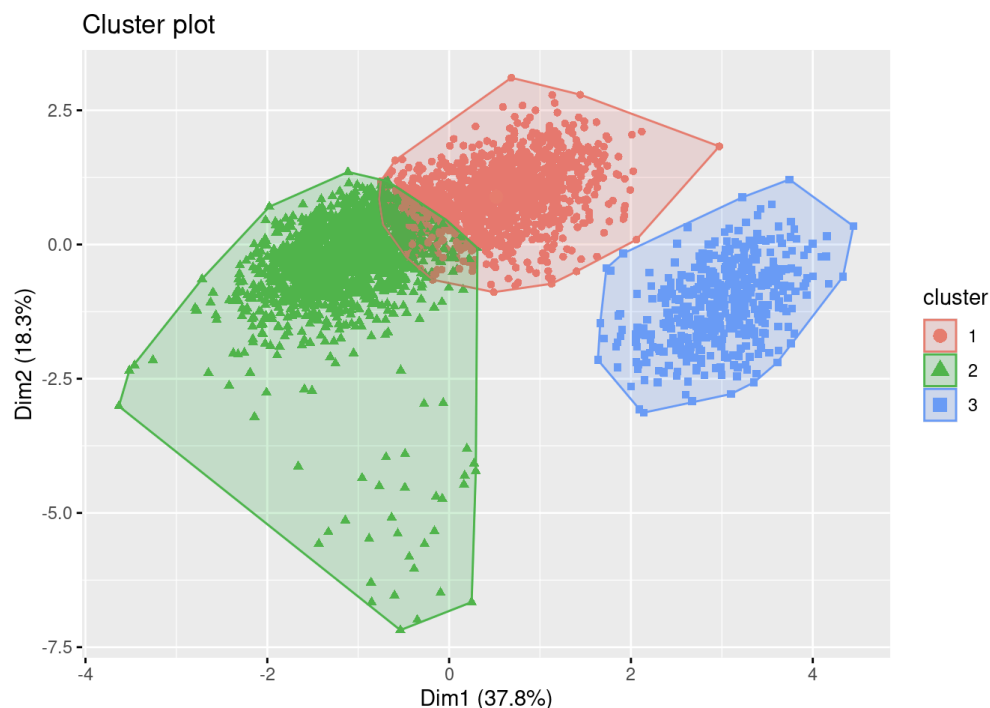
K-means Clustering

K-means clustering is an unsupervised machine learning algorithm used to partition a dataset into K distinct, non-overlapping subgroups (clusters). The goal of K-means is to group data points such that the points in the same cluster are more similar to each other than to those in other clusters. For this analysis, K-means clustering was applied to classify Kershaw's pitches into different clusters based on their characteristics.

Steps for K-means Clustering:

1. **Feature Selection:** The relevant features for clustering were selected, including release speed, release spin rate, horizontal movement (pfx_x), vertical movement (pfx_z), launch angle, and launch speed.
2. **Data Scaling:** The selected features were scaled to ensure that each feature contributes equally to the clustering process.
3. **Optimal Number of Clusters:** The optimal number of clusters (K) was determined using the elbow method, which involves plotting the within-cluster sum of squares (WCSS) against the number of clusters and looking for the "elbow point" where the rate of decrease slows down.
4. **K-means Clustering:** The K-means algorithm was applied with the determined number of clusters to partition the data.
5. **Cluster Assignment:** Each pitch was assigned to one of the clusters based on its characteristics.
6. **Cluster Analysis:** The characteristics and effectiveness of each cluster were analyzed to provide insights into Kershaw's pitching performance.

After clustering, each pitch was assigned to one of the three clusters based on its characteristics. The clusters were then analyzed to understand their profiles:



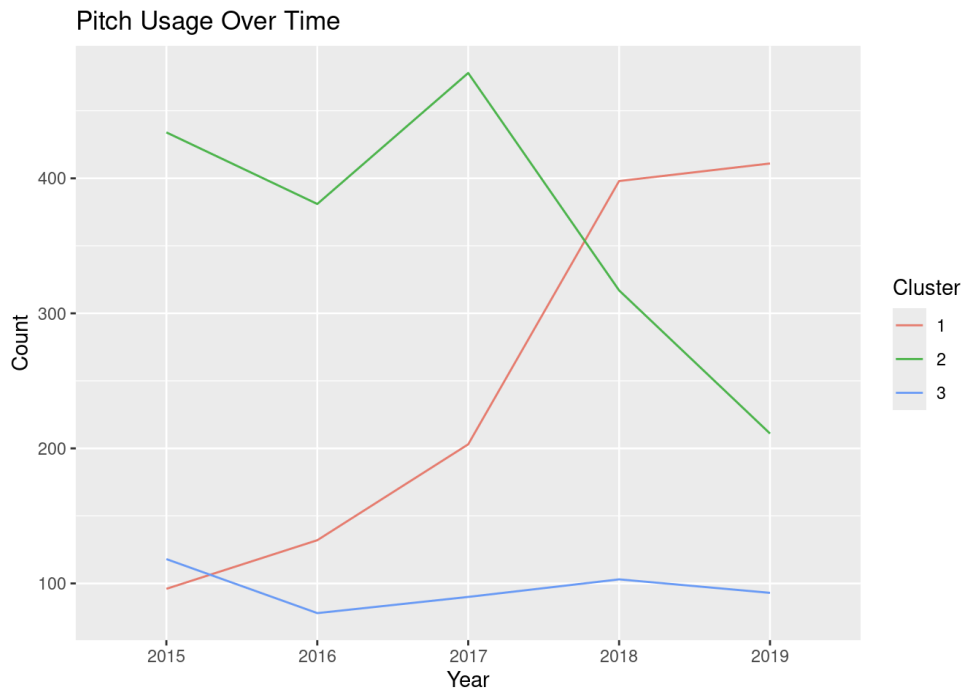
Cluster Analysis

- **Cluster 1:** Represented pitches with balanced speed and spin rate, often resulting in moderate effectiveness.
 - **Release Speed:** Mean = 88.6 mph
 - **Spin Rate:** Mean = 2541 rpm
 - **Horizontal Movement (pfx_x):** Mean = -0.6

- **Vertical Movement (pfx_z):** Mean = 5.8
- **Launch Angle:** Mean = -0.2 degrees
- **Cluster 2:** High-speed pitches, which were more likely to be effective in avoiding hits.
 - **Release Speed:** Mean = 92.4 mph
 - **Spin Rate:** Mean = 2313 rpm
 - **Horizontal Movement (pfx_x):** Mean = 0.2
 - **Vertical Movement (pfx_z):** Mean = 6.4
 - **Launch Angle:** Mean = 1.5 degrees
- **Cluster 3:** Slower pitches with significant movement, typically curveballs.
 - **Release Speed:** Mean = 73.5 mph
 - **Spin Rate:** Mean = 2368 rpm
 - **Horizontal Movement (pfx_x):** Mean = -0.8
 - **Vertical Movement (pfx_z):** Mean = -0.4
 - **Launch Angle:** Mean = -2.3 degrees

Pitch Usage Over Time

The usage of different pitch types was analyzed over the years to understand changes in Kershaw's pitching strategy. The count of pitches in each cluster was plotted against the year to visualize trends.



Pitch Effectiveness

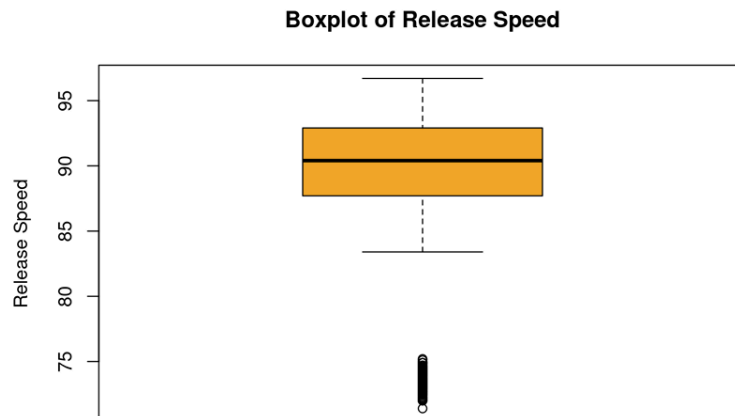
The effectiveness of each pitch type was evaluated based on hit rates. The number of hits (singles, doubles, triples, home runs) for each cluster was calculated, and the hit rate was determined by dividing the number of hits by the total number of pitches in each cluster.

- **Cluster 1:** Hit rate = 21.1%
- **Cluster 2:** Hit rate = 19.4%
- **Cluster 3:** Hit rate = 20.1%

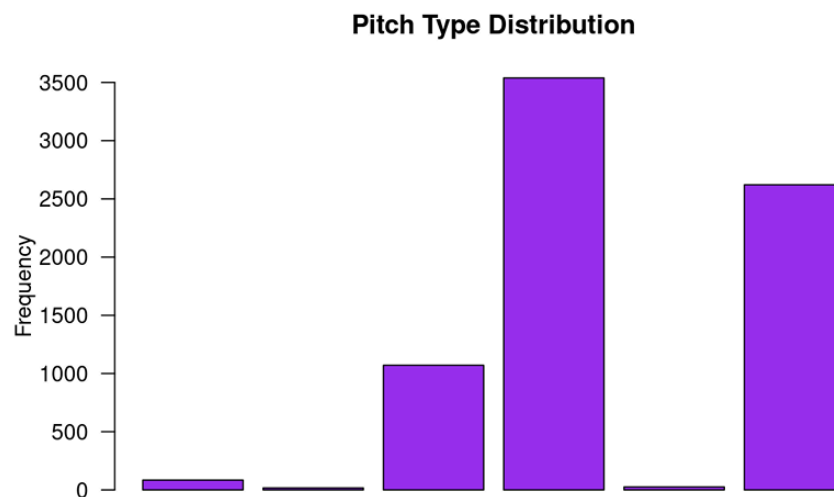
Visualization

The following visualizations were created to illustrate the findings:

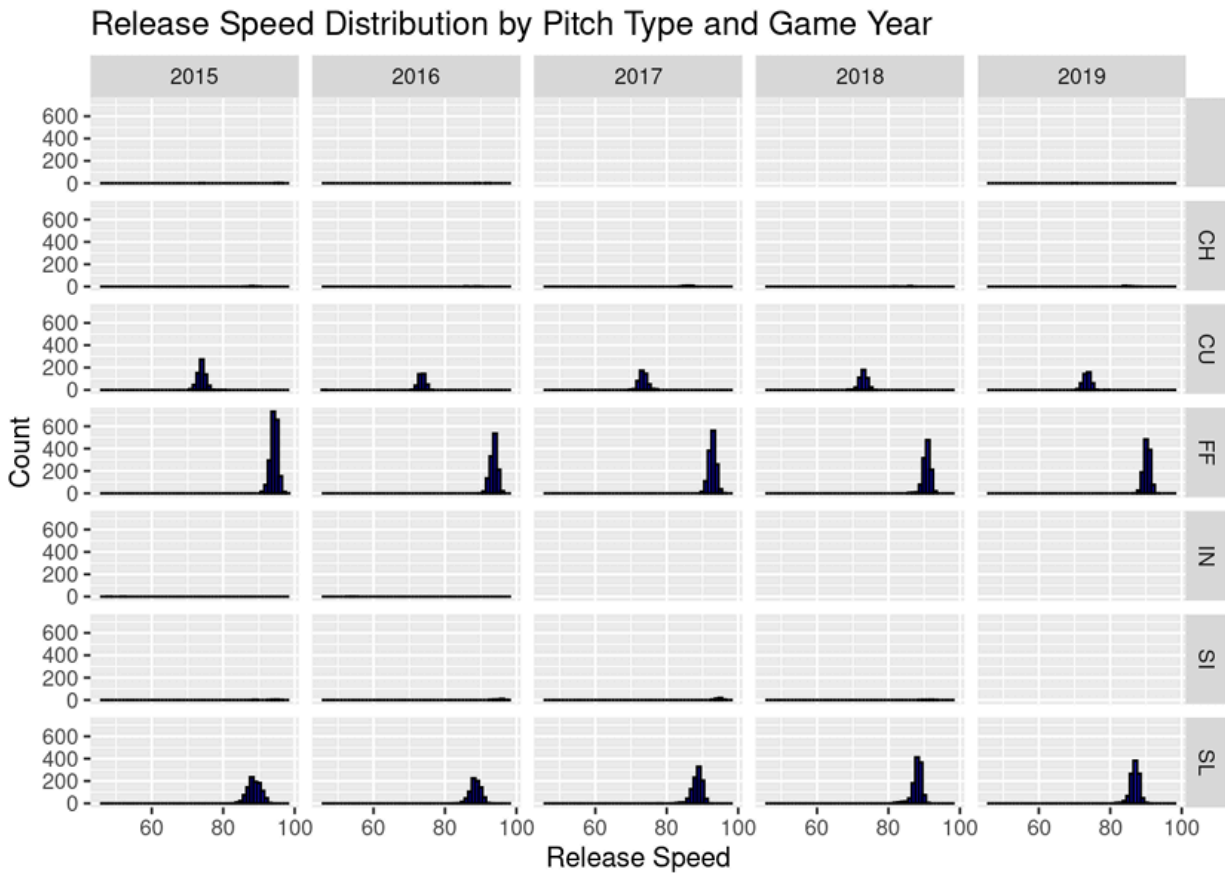
1. **Boxplot of Release Speed:** Displayed the distribution of release speeds for all pitches, highlighting the central tendency and variability.
2. **Bar Plot of Pitch Type Distribution:** Showed the frequency of different pitch types, emphasizing the predominance of Four-Seam Fastballs and Sliders.
3. **Histogram of Release Speed:** Illustrated the distribution of release speeds by pitch type and year, revealing trends and patterns over time.
4. **Loess Smoothing Plots:** Depicted trends in release speed and movement over time, offering insights into seasonal and yearly changes.



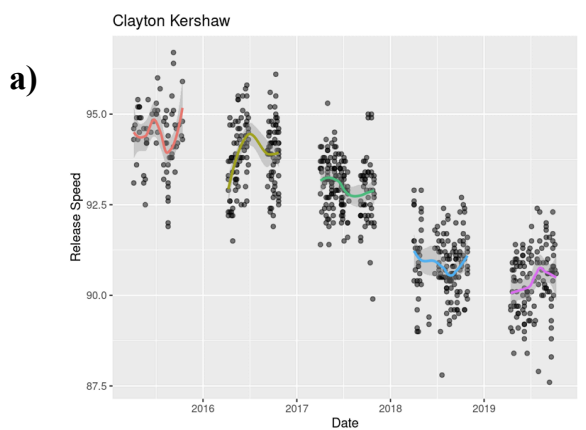
Here we can see a distribution of all pitches Kershaw threw in the 5 year span from 2015-2019. The median release speed seems to be approximately 90 mph.



When it comes to strikes, it seems that Kershaw's favorite pitches are fastballs, sliders, and curveballs.

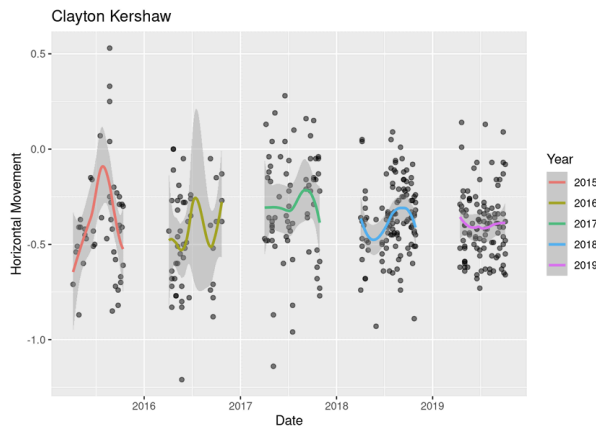


This displays that as we go from 2015-2019, we see a slight decrease in the speed of the FF pitches. In addition, Kershaw relies more on the slider and less on the fastball as the years pass.



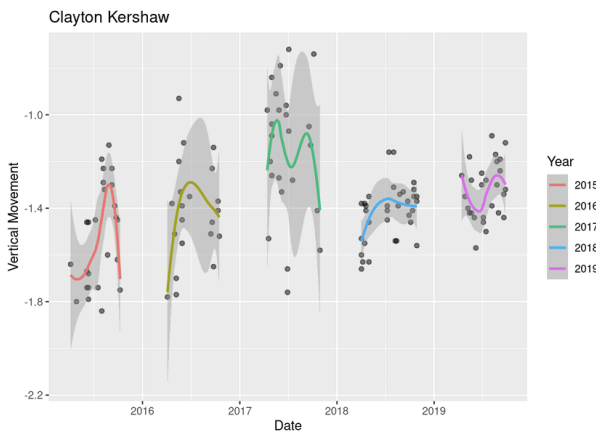
Seam Fastball is a pitch that is best if it is as fast as possible. Therefore, we can see a clear decline through the years of Kershaw's career. When it comes to each individual season, there is no clear pattern of it decreasing as the season goes on (which is surprising).

b)



Slider pitches rely on horizontal movement to be effective. The higher the magnitude of movement, the better. We can see that the 2015 season had some of Kershaw's highest movement sliders but also some of his lowest, while the 2019 season maintained a leveled magnitude throughout the season. Different from his fast ball performances, there is no year by year trend of his performance decreasing when it comes to the slider.

c)

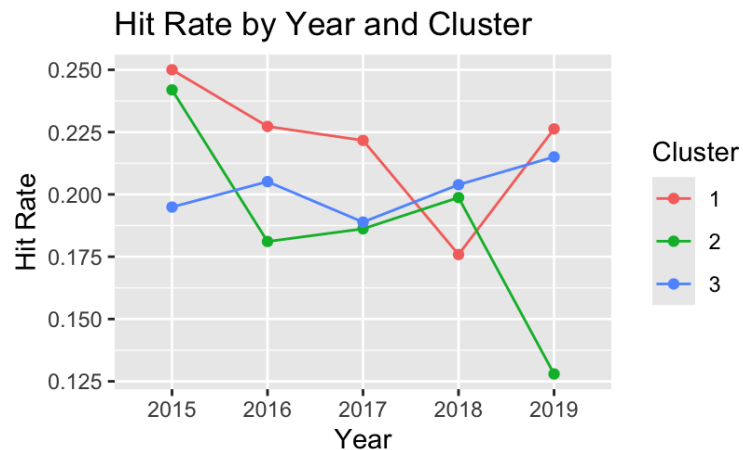


Curveball pitches rely on vertical movement to be effective. The higher the magnitude of movement, the better. We can see the level of vertical movement in his curveball pitches is highest in 2015, and it slowly begins to decrease from there. The 2017 season seemed the lowest level in this case. The level of decrease in performance in his curveball from 2016 to 2017 is surprising. This could be due to injury or etc.

Result and Discussion

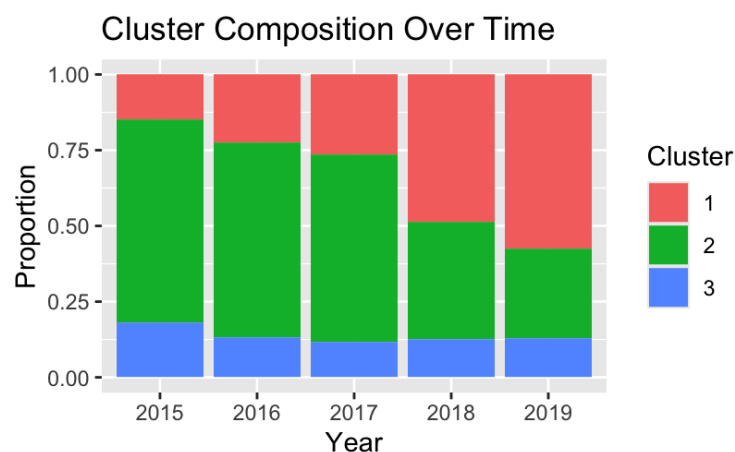
We conducted the analysis of Clayton Kershaw's pitching performance over the five years period from 2015 to 2019 through k-mean clustering method. We categorized Kershaw's pitches into three distinct clusters based on their release speed, horizontal and vertical movement, and launch angle. The effectiveness of each cluster was assessed using hit rates, where a lower hit rate indicates a better performance. The evolution of pitch usage over time was also examined.

The hit rates for each cluster were analyzed over the five years:



According to the line graph, the hit rate for pitches in cluster 1 decreased from 2015 to 2018, then increased from 2018 to 2019. The initial decrease in hit rates from 2015 to 2018 suggests that these pitches became more effective during this period. However, the increase in 2019 indicates a potential decline in effectiveness. The hit rate for this cluster is 21.1%, indicating this type of pitches are generally more hittable and less effective. For cluster 2, the generally decreasing hit rate trend highlights its increasing effectiveness over time. It has the lowest hit rate of 19.4%, suggesting that the pitches characterized by high release speed and strong vertical movement are the most effective pitch type for Clayton Kershaw. Cluster 3 has a relatively stable and low hit rate of 20.1%, indicating that these pitches are consistently effective.

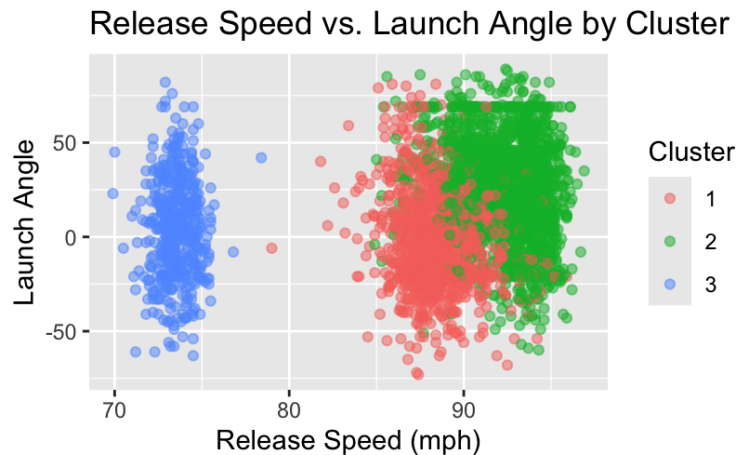
The composition of pitches over time:



The proportion of pitches in cluster 1 gradually increased from 2015 to 2019, indicating that Kershaw had a growing reliance on this type of pitch. However, the effectiveness of this increasing reliance on cluster 1 pitches should be evaluated, since it may not yield the desired results due to the higher hit rates. The proportion of cluster 2 pitches has decreased over the same period, despite these pitches having been the most effective. The less frequent use of cluster 2 pitches may suggest potential physical challenges of

performing these high speed and low spin pitches. The proportion of cluster 3 pitches remains low and relatively stable through the 5 years, suggesting a less frequent but consistent use of these low speed and moderate spin pitches with a downward break.

The relationship between release speed and launch angle for each cluster:



According to the plot, cluster 1 pitches have moderate speeds and are concentrated around zero launch angles, suggesting that these pitches are frequently hit but not for significant power. The moderate release speeds and close-to-zero launch angles potentially make them more predictable and easier to hit for batters, which explains the higher hit rates. Cluster 2 pitches have highest release speeds and a larger proportion of launch angles above 50 degrees, suggesting they are harder to hit and may lead to fly-balls. The high release speed and stronger vertical movement contribute to their lowest hit rate. Cluster 3 pitches have distinctly lower release speeds and varied launch angles, indicating these pitches induce weaker contact that may lead to ground balls or pop-ups. The low release speeds and varied launch angles contribute to their lower hit rate.

Conclusion

The analysis revealed several key insights into Clayton Kershaw's pitch performance over a 5 year span:

- There is a noticeable decline in the release speed of his pitches over the years, particularly for the Four-Seam Fastball.
- The horizontal movement of his Slider remained relatively consistent, indicating effective control.
- The vertical movement of his Curveball showed a decline, suggesting a reduction in effectiveness over time.
- K-means clustering provided a useful framework for classifying pitches and understanding their characteristics and effectiveness.
- Cluster 2 pitches, characterized by high release speed and significant vertical movement, were the most effective in avoiding hits.

We conclude that Clayton Kershaw is an elite pitcher with 3 distinct pitches he throws the most frequently: slider, four-seam fastball, and curveball. Each of these 3 pitches are very solid as their average

hit rate is 20%. In terms of effectiveness over 2015-2019, the Hit Rate by Year and Cluster visual shows a minor difference in average hit rate of all pitches by 3-4% meaning towards the end of the 5 year gap we analyzed, Kershaw actually became more effective. Lastly, towards the end of our analysis interval, Kershaw focused more on his offspeed pitches, rather than high velocity; which is what made his fastball more remarkable and effective.

Appendix

```
library(baseballr)
library(dplyr)
library(ggplot2)
library(reshape2)
library(zoo)
```

```
player_id <- playerid_lookup("Kershaw") %>%
  dplyr::filter(first_name == "Clayton") %>%
  dplyr::select(mlbam_id, first_name, last_name)
player_id[1,1]
```

```
# 477132
# updated functions in the package `baseballr`
```

```
csv_from_url <- function(...){
  data.table::fread(...)
}
```

```
##
```

```
make_baseballr_data <- function(df, type, timestamp){
  out <- df %>%
    tidyr::as_tibble()

  class(out) <- c("baseballr_data", "tbl_df", "tbl", "data.table", "data.frame")
  attr(out, "baseballr_timestamp") <- timestamp
  attr(out, "baseballr_type") <- type
  return(out)
}
```

```
# @title
statcast_search <- function(start_date = Sys.Date() - 1, end_date = Sys.Date(),
                             playerid = NULL,
                             player_type = "batter", ...) {
  # Check for other user errors.
  if (start_date <= "2015-03-01") { # March 1, 2015 was the first date of Spring
    Training.
    message("Some metrics such as Exit Velocity and Batted Ball Events have only
    been compiled since 2015.")
  }
  if (start_date < "2008-03-25") { # March 25, 2008 was the first date of the 2008
    season.
    stop("The data are limited to the 2008 MLB season and after.")
    return(NULL)
  }
}
```

```

if (start_date == Sys.Date()) {
  message("The data are collected daily at 3 a.m. Some of today's games may not
be included.")
}
if (start_date > as.Date(end_date)) {
  stop("The start date is later than the end date.")
  return(NULL)
}

```

```

playerid_var <- ifelse(player_type == "pitcher",
  "pitchers_lookup%5B%5D", "batters_lookup%5B%5D")

vars <- tibble::tribble(
  ~var, ~value,
  "all", "true",
  "hfPT", "",
  "hfAB", "",
  "hfBBT", "",
  "hfPR", "",
  "hfZ", "",
  "stadium", "",
  "hfBBL", "",
  "hfNewZones", "",
  "hfGT", "R%7CP0%7CS%7C&hfC",
  "hfSea", paste0(lubridate::year(start_date), "%7C"),
  "hfSit", "",
  "hfOuts", "",
  "opponent", "",
  "pitcher_throws", "",
  "batter_stands", "",
  "hfSA", "",
  "player_type", player_type,
  "hfInfield", "",
  "team", "",
  "position", "",
  "hfOutfield", "",
  "hfRO", "",
  "home_road", "",
  playerid_var, ifelse(is.null(playerid), "", as.character(playerid)),
  "game_date_gt", as.character(start_date),
  "game_date_lt", as.character(end_date),
  "hfFlag", "",
  "hfPull", "",
  "metric_1", "",
  "hfInn", "",
  "min_pitches", "0",

```

```

"min_results", "0",
"group_by", "name",
"sort_col", "pitches",
"player_event_sort", "h_launch_speed",
"sort_order", "desc",
"min_abs", "0",
"type", "details") %>%
dplyr::mutate(pairs = paste0(.data$var, "=", .data$value))

```

```

if (is.null(playerid)) {
  # message("No playerid specified. Collecting data for all batters/pitchers.")
  vars <- vars %>%
    dplyr::filter(!grepl("lookup", .data$var))
}

```

```

url_vars <- paste0(vars$pairs, collapse = "&")
url <- paste0("https://baseballsavant.mlb.com/statcast_search/csv?", url_vars)
# message(url)

```

```

# Do a try/catch to show errors that the user may encounter while downloading.
tryCatch(
  {
    suppressMessages(
      suppressWarnings(
        payload <- csv_from_url(url, encoding = "UTF-8")
      )
    )
  },
  error = function(cond) {
    message(cond)
    stop("No payload acquired")
  },
  # this will never run??
  warning = function(cond) {
    message(cond)
  }
)

```

```

# returns 0 rows on failure but > 1 columns
if (nrow(payload) > 1) {

  names(payload) <- c("pitch_type", "game_date", "release_speed",
"release_pos_x",
                    "release_pos_z", "player_name", "batter", "pitcher",

```

```

"events",
      "description", "spin_dir", "spin_rate_deprecated",
"break_angle_deprecated",
      "break_length_deprecated", "zone", "des", "game_type",
"stand",
      "p_throws", "home_team", "away_team", "type",
"hit_location",
      "bb_type", "balls", "strikes", "game_year", "pfx_x",
"pfx_z",
      "plate_x", "plate_z", "on_3b", "on_2b", "on_1b",
"outs_when_up",
      "inning", "inning_topbot", "hc_x", "hc_y",
"tfs_deprecated",
      "tfs_zulu_deprecated", "fielder_2", "umpire", "sv_id",
"vx0",
      "vy0", "vz0", "ax", "ay", "az", "sz_top", "sz_bot",
"hit_distance_sc",
      "launch_speed", "launch_angle", "effective_speed",
"release_spin_rate",
      "release_extension", "game_pk", "pitcher_1", "fielder_2_1",
      "fielder_3", "fielder_4", "fielder_5", "fielder_6",
"fielder_7",
      "fielder_8", "fielder_9", "release_pos_y",
"estimated_ba_using_speedangle",
      "estimated_woba_using_speedangle", "woba_value",
"woba_denom",
      "babip_value", "iso_value", "launch_speed_angle",
"at_bat_number",
      "pitch_number", "pitch_name", "home_score", "away_score",
"bat_score",
      "fld_score", "post_away_score", "post_home_score",
"post_bat_score",
      "post_fld_score", "if_fielding_alignment",
"of_fielding_alignment",
      "spin_axis", "delta_home_win_exp", "delta_run_exp",
"bat_speed", "swing_length")
  payload <- process_statcast_payload(payload) %>%
    make_baseballr_data("MLB Baseball Savant Statcast Search data from
baseballsavant.mlb.com", Sys.time())
  return(payload)
} else {
  warning("No valid data found")
}

# (somewhere within the statcast_search function before the payload is searched
for)
colos <- c("pitch_type", "game_date",

```

```

"release_speed", "release_pos_x", "release_pos_z",
"player_name", "batter", "pitcher",
"events", "description", "spin_dir",
"spin_rate_deprecated", "break_angle_deprecated",
"break_length_deprecated", "zone", "des",
"game_type", "stand", "p_throws",
"home_team", "away_team", "type",
"hit_location", "bb_type", "balls",
"strikes", "game_year", "pfx_x",
"pfx_z", "plate_x", "plate_z",
"on_3b", "on_2b", "on_1b", "outs_when_up",
"inning", "inning_topbot", "hc_x",
"hc_y", "tfs_deprecated", "tfs_zulu_deprecated",
"fielder_2", "umpire", "sv_id",
"vx0", "vy0", "vz0", "ax",
"ay", "az", "sz_top", "sz_bot",
"hit_distance_sc", "launch_speed", "launch_angle",
"effective_speed", "release_spin_rate",
"release_extension", "game_pk", "pitcher_1",
"fielder_2_1", "fielder_3", "fielder_4",
"fielder_5", "fielder_6", "fielder_7",
"fielder_8", "fielder_9", "release_pos_y",
"estimated_ba_using_speedangle", "estimated_woba_using_speedangle",
"woba_value", "woba_denom", "babip_value",
"iso_value", "launch_speed_angle", "at_bat_number",
"pitch_number", "pitch_name", "home_score",
"away_score", "bat_score", "fld_score",
"post_away_score", "post_home_score",
"post_bat_score", "post_fld_score", "if_fielding_alignment",
"of_fielding_alignment", "spin_axis",
"delta_home_win_exp", "delta_run_exp")
colNumber <- ncol(payload)

```

```

if(length(colos) != colNumber){
  newCols <- paste("newStat", 1:(length(colos) - colNumber))
  colos <- c(colos, newCols)
  message("New stats detected! baseballr will be updated soon to properly identify
these stats")
}
# payload is acquired somewhere in here
# when the payload columns need to be named:
names(payload) <- colos

```

```

payload <- payload %>%
  make_baseballr_data("MLB Baseball Savant Statcast Search data from
baseballsavant.mlb.com", Sys.time())
  return(payload)
}
}

```

```

statcast_search.default <- function(start_date = Sys.Date() - 1, end_date =
Sys.Date(),
                                   playerid = NULL, player_type =
"batter", ...) {

  message(paste0(start_date, " is not a date. Attempting to coerce..."))
  start_Date <- as.Date(start_date)

```

```

  tryCatch(
    {
      end_Date <- as.Date(end_date)
    },
    warning = function(cond) {
      message(paste0(end_date, " was not coercible into a date. Using today. "))
      end_Date <- Sys.Date()
      message("Original warning message:")
      message(cond)
    }
  )

  statcast_search(start_Date, end_Date,
                  playerid, player_type, ...)
}

statcast_search_batters <- function(start_date, end_date, batterid = NULL, ...) {
  statcast_search(start_date, end_date, playerid = batterid,
                  player_type = "batter", ...)
}

statcast_search_pitchers <- function(start_date, end_date, pitcherid = NULL, ...) {
  statcast_search(start_date, end_date, playerid = pitcherid,
                  player_type = "pitcher", ...)
}

player_15 <- statcast_search(start_date = "2015-03-31", end_date = "2015-10-31",
                             playerid = 477132, player_type = 'pitcher')
player_16 <- statcast_search(start_date = "2016-03-31", end_date = "2016-10-31",

```

```

playerid = 477132, player_type = 'pitcher')
player_17 <- statcast_search(start_date ="2017-03-31", end_date ="2017-10-31",
playerid = 477132, player_type = 'pitcher')
player_18 <- statcast_search(start_date ="2018-03-31", end_date ="2018-10-31",
playerid = 477132, player_type = 'pitcher')
player_19 <- statcast_search(start_date ="2019-03-31", end_date ="2019-10-31",
playerid = 477132, player_type = 'pitcher')

```

```

# type  ball or strike or ball in play
player1 <- bind_rows(player_15,player_16,player_17,player_18,player_19) %>%
  mutate(Year = as.factor(substr(game_date,1,4))) %>%
  filter(type == "S") %>%
  filter(launch_speed != 0)

# type  strike
player2 <- bind_rows(player_15,player_16,player_17,player_18,player_19) %>%
  mutate(Year = as.factor(substr(game_date,1,4))) %>%
  filter(type == "S")

# type  ball in play only
player3 <- bind_rows(player_15,player_16,player_17,player_18,player_19) %>%
  mutate(Year = as.factor(substr(game_date,1,4))) %>%
  filter(launch_speed != 0)

# any pitch
player4 <- bind_rows(player_15,player_16,player_17,player_18,player_19) %>%
  mutate(Year = as.factor(substr(game_date,1,4)))

# column names
print(colnames(player1))

# Summary statistics for a release speed
summary(player4$release_speed)

# Summary statistics for a pitch type
table(player4$pitch_type)
boxplot(player1$release_speed, main="Boxplot of Release Speed", ylab="Release
Speed", col="orange")

barplot(table(player4$pitch_type), main="Pitch Type Distribution", xlab="Pitch
Type", ylab="Frequency", col="purple", las=2)

# Summary statistics for a release speed
summary(player3$release_speed)

# Summary statistics for a pitch type
table(player3$pitch_type)

```

```
# Summary statistics for a release speed
summary(player2$release_speed)
```

```
# Summary statistics for a pitch type
table(player2$pitch_type)
barplot(table(player3$pitch_type), main="Pitch Type Distribution", xlab="Pitch
Type", ylab="Frequency", col="purple", las=2)
barplot(table(player2$pitch_type), main="Pitch Type Distribution", xlab="Pitch
Type", ylab="Frequency", col="purple", las=2)
library(ggplot2)

ggplot(player4, aes(x=release_speed)) +
  geom_histogram(binwidth=1, fill="blue", color="black") +
  facet_grid(pitch_type ~ game_year) +
  labs(title="Release Speed Distribution by Pitch Type and Game Year", x="Release
Speed", y="Count")
player_avg_yr <- player1 %>%
  group_by(Year) %>%
  summarise(
    angle = round(mean(launch_angle, na.rm = TRUE),1),
    speed = round(mean(launch_speed, na.rm = TRUE),1))

player_avg_yr
player1 %>%
  ggplot(aes(game_date, launch_speed)) +
  geom_point(alpha = 0.5) +
  stat_smooth(aes(group = Year, color = Year), formula = y~x, method='loess')+
  ggtitle("Clayton Kershaw") +
  ylab("Speed = MPH") +
  xlab("Date")

player1 %>%
  filter(pitch_type=="FF") %>%
  ggplot(aes(game_date, release_speed)) +
  geom_point(alpha = 0.5) +
  stat_smooth(aes(group = Year, color = Year), formula = y~x, method='loess')+
  ggtitle("Clayton Kershaw") +
  ylab("Release Speed") +
  xlab("Date")

player1 %>%
  filter(pitch_type=="SL") %>%
  ggplot(aes(game_date, pfx_x)) +
  geom_point(alpha = 0.5) +
  stat_smooth(aes(group = Year, color = Year), formula = y~x, method='loess')+
  ggtitle("Clayton Kershaw") +
  ylab("Horizontal Movement") +
```

```

  xlab("Date")
player1 %>%
  filter(pitch_type=="CU") %>%
  ggplot(aes(game_date, pfx_z)) +
  geom_point(alpha = 0.5) +
  stat_smooth(aes(group = Year, color = Year), formula = y~x, method='loess')+
  ggtitle("Clayton Kershaw") +
  ylab("Vertical Movement") +
  xlab("Date")

```

```

# Load necessary libraries
library(dplyr)
library(ggplot2)
library(factoextra)

# Assuming player4 contains the complete data for the years 2015-2019
# Create the Year column
player4 <- player4 %>%
  mutate(Year = as.factor(format(as.Date(game_date), "%Y")))

# Select relevant features, including the 'events' column
pitch_data <- player4 %>%
  select(release_speed, release_spin_rate, pfx_x, pfx_z, launch_angle,
  launch_speed, Year, events) %>%
  na.omit() # Remove rows with missing values

# Scale the data, excluding non-numeric columns
pitch_data_scaled <- scale(pitch_data %>% select(release_speed, release_spin_rate,
pfx_x, pfx_z, launch_angle, launch_speed))

# Perform K-means clustering with the optimal number of clusters (3)
set.seed(123)
kmeans_result <- kmeans(pitch_data_scaled, centers = 3, nstart = 25)

# Add cluster assignment to the original data
pitch_data <- cbind(pitch_data, cluster = kmeans_result$cluster)

# Visualize the clusters
fviz_cluster(kmeans_result, data = pitch_data_scaled, geom = "point", ellipse.type
= "convex")

# Create detailed profiles for each cluster
cluster_profiles <- pitch_data %>%
  group_by(cluster) %>%
  summarise(
    release_speed_mean = mean(release_speed),
    release_speed_sd = sd(release_speed),

```

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    release_spin_rate_mean = mean(release_spin_rate),
    release_spin_rate_sd = sd(release_spin_rate),
    pfx_x_mean = mean(pfx_x),
    pfx_x_sd = sd(pfx_x),
    pfx_z_mean = mean(pfx_z),
    pfx_z_sd = sd(pfx_z),
    launch_angle_mean = mean(launch_angle),
    launch_angle_sd = sd(launch_angle),
    launch_speed_mean = mean(launch_speed),
    launch_speed_sd = sd(launch_speed)
  )

print(cluster_profiles)
# Visualize the distribution of release speed for each cluster
ggplot(pitch_data, aes(x = release_speed, fill = factor(cluster))) +
  geom_density(alpha = 0.5) +
  labs(title = "Distribution of Release Speed by Cluster", x = "Release Speed
(mph)", fill = "Cluster")
# Plot pitch usage over time
pitch_data %>%
  group_by(Year, cluster) %>%
  summarise(count = n()) %>%
  ggplot(aes(x = Year, y = count, color = factor(cluster), group = cluster)) +
  geom_line() +
  labs(title = "Pitch Usage Over Time", x = "Year", y = "Count", color = "Cluster")
# Evaluate pitch effectiveness based on hits (single, double, triple, home_run)
effectiveness <- pitch_data %>%
  group_by(cluster) %>%
  summarise(
    hits = sum(events %in% c("single", "double", "triple", "home_run")),
    total_pitches = n(),
    hit_rate = hits / total_pitches
  )

print(effectiveness)

```

```

# Additional Visualizations

# Hit rate by year and cluster
effectiveness_by_year <- pitch_data %>%
  group_by(Year, cluster) %>%
  summarise(
    hits = sum(events %in% c("single", "double", "triple", "home_run")),
    total_pitches = n(),
    hit_rate = hits / total_pitches
  )

```

```

)

ggplot(effectiveness_by_year, aes(x = Year, y = hit_rate, color =
factor(cluster), group = cluster)) +
  geom_line() +
  geom_point() +
  labs(title = "Hit Rate by Year and Cluster", x = "Year", y = "Hit Rate",
color = "Cluster")

# Cluster composition over time
cluster_composition <- pitch_data %>%
  group_by(Year, cluster) %>%
  summarise(count = n()) %>%
  mutate(proportion = count / sum(count))

ggplot(cluster_composition, aes(x = Year, y = proportion, fill =
factor(cluster))) +
  geom_bar(stat = "identity", position = "fill") +
  labs(title = "Cluster Composition Over Time", x = "Year", y =
"Proportion", fill = "Cluster")

# Scatter plot of release speed vs. launch angle by cluster
ggplot(pitch_data, aes(x = release_speed, y = launch_angle, color =
factor(cluster))) +
  geom_point(alpha = 0.5) +
  labs(title = "Release Speed vs. Launch Angle by Cluster", x = "Release
Speed (mph)", y = "Launch Angle", color = "Cluster")

```